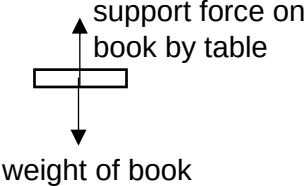
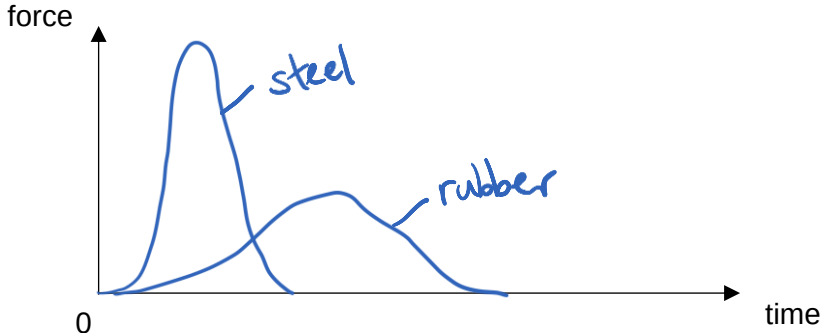


1biv 2.	gravitational potential energy to thermal/internal energy	B1
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2ai	 support force on book by table weight of book Either The support force on book by table and the weight of book are both acting on the book. (N3L states that forces act on different bodies.) They are different types of forces, <u>gravitational</u> and <u>electromagnetic/contact</u> forces. (N3L states that forces must be of the same type.) (So, they are not a pair of action-reaction forces.) Or Reaction force of contact force by table on book is <u>contact</u> force that book exerts on table. Reaction force of weight of book is <u>gravitational</u> force that book exerts on Earth.	B1 B1 B1 B1
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	(So, weight of book and contact force by table on book are not a pair of action-reaction forces.) Credit 1 mark only if student did not mention type of force.	
2a ii	Since book is resting on table, there is no net force on the book, so the two forces are equal and opposite.	B1
2b	Suppose two colliding bodies A and B (where A and B is an isolated system), By Newton's third law, force A exert on B, F_{AB} is equal in magnitude and opposite in direction to force B exert on A, F_{BA}. $F_{AB} = -F_{BA}$ Duration of collision is the same for A and B. By Newton's second law, net force on A, F_{BA} is equal to rate of change of momentum of A. Net force on B, F_{AB} is equal to rate of change of momentum of B. Hence, total (rate of) change of momentum is 0.	B1 B1 B1
2c	By Conservation of Linear Momentum Sum of initial momentum = Sum of final momentum $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ $(28)(88) + (17)(53) = (28)(67) + (17)v_2$ $v_2 = 87.6 \text{ m s}^{-1}$ Loss in kinetic energy = total initial kinetic energy – total final kinetic energy $= \frac{1}{2} (28)(88)^2 + \frac{1}{2} (17)(53)^2 - (\frac{1}{2} (28)(67)^2 + \frac{1}{2} (17)(87.6)^2)$ $= 4200 \text{ J}$	C1 C1 A1
2d	 Correct shape – smooth curves for both lines. Line for steel objects has larger peak and smaller duration than line for rubber, with approximately equal area under the two lines.	B1 B1